

Dependency Injection

It is a design pattern that implements inversion of control in order to resolve dependencies.

Dependency is an object which can be used as a service. Injection is nothing but passing dependency to a object (dependent) that would use it. The dependency injection is technique for keeping the code modular and least coupled. It's a generic design pattern which can be applied for most of the programming languages. The angular framework also has dependency injection feature built in.

Dependency Injection

"Dependency Injection" is a 25-dollar term for a 5-cent concept.

A popular saying about the concept of dependency injection

Why dependency injection,

Consider the following example class in typescript,

```

• export class Bike {
•   public engine: BikeEngine;
•   public tires: BikeTires;
•   public description = 'No DI';
•
•   constructor() {
•     this.engine = new BikeEngine();
•     this.tires = new BikeTires(); }
•
•   public ride() {
•     return `${this.description} bike with ${this.
engine.cylinders} cylinders and ${this.tires.make} tires.`;}}

```

Above class creates its dependencies inside its constructor. This makes dependency classes BikeEngine and BikeTire very difficult to change. It also uses very specific classes, if in future we needed different type of engine,